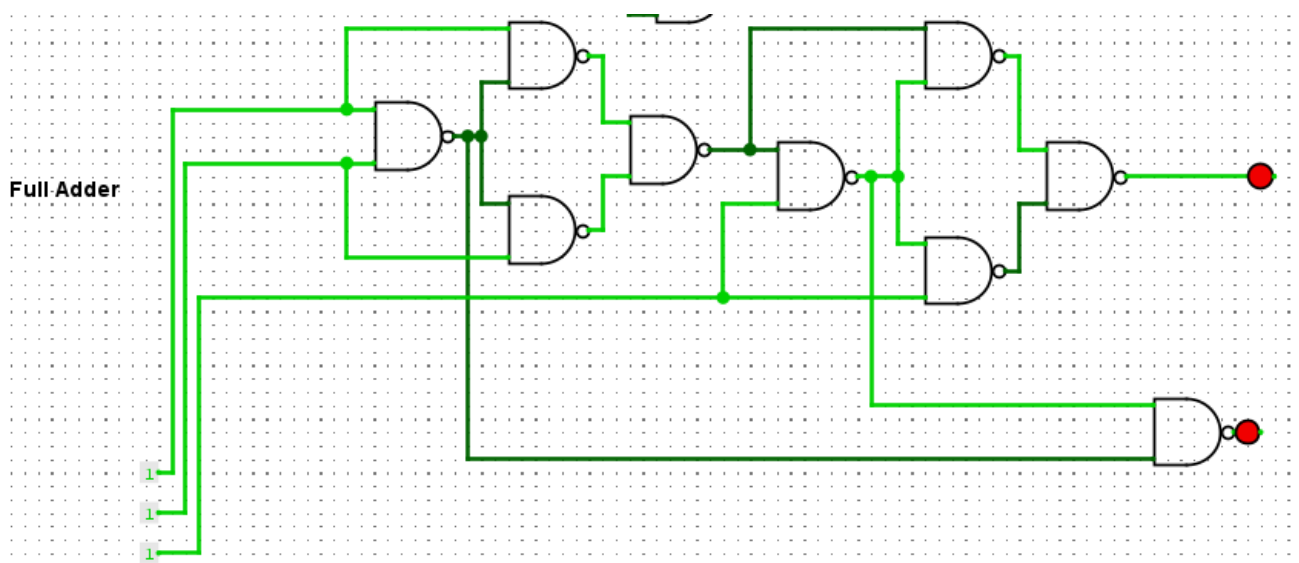
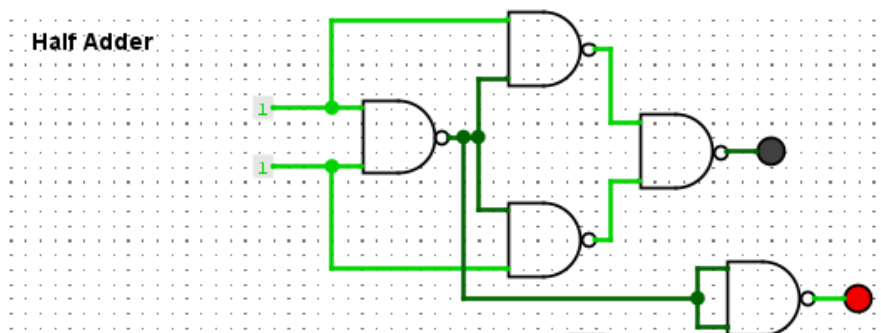
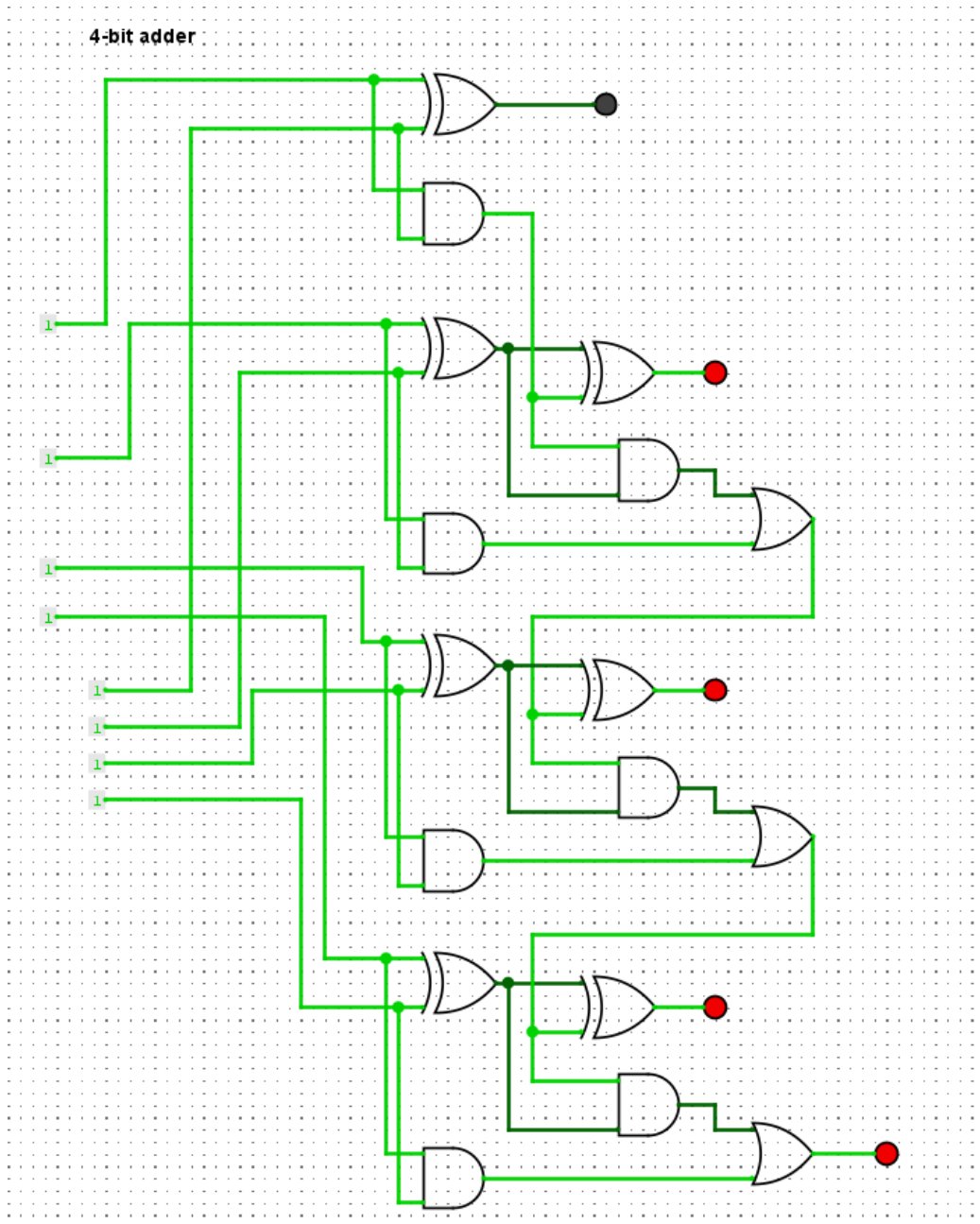


Full Adder

Procedure:

1. Open Logisim and create a new project.
2. Design the **Half Adder** using one XOR gate (for Sum) and one AND gate (for Carry). Connect inputs A and B and verify outputs.
3. Design the **Full Adder** using two Half Adders and an OR gate, with inputs A, B, and Cin.
4. Design the **4-bit Adder** by cascading four Full Adder modules. Connect the Cout of each stage to the Cin of the next.
5. Apply input combinations to each circuit and observe the output LEDs.
6. Record observations and verify against the expected truth table.





Conclusion:

The Half Adder, Full Adder, and 4-bit Adder circuits were successfully designed and simulated in Logisim. The outputs matched the expected Boolean expressions and truth tables for all input combinations. It was observed that the Carry-out in the 4-bit adder becomes 1 when the result exceeds 15, indicating overflow. The experiment demonstrates how complex arithmetic circuits are built by cascading simpler logic gate modules.